# Biology-1: Lecture 04

## (BIO103)

## Chapter 1. Introduction to Biology

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# History of Biology

**Lecture 4: Outlines** **Spring, 2026**

## 1. History of biology

# History of Biology

- The history of biology traces the study of the living world from ancient to modern times.

- The biological sciences emerged from **traditions of medicine** and **natural history** reaching back to Ayurveda, ancient Egyptian medicine and the works of **Aristotle and Galen** in the ancient Greco-Roman world.

- This ancient work was further developed in the **Middle Ages by Muslim physicians** and scholars such as **Avicenna**.

**Aristotle-an Ancient Greek philosopher and polymath**

Galen-an Ancient Greek philosopher and biologist

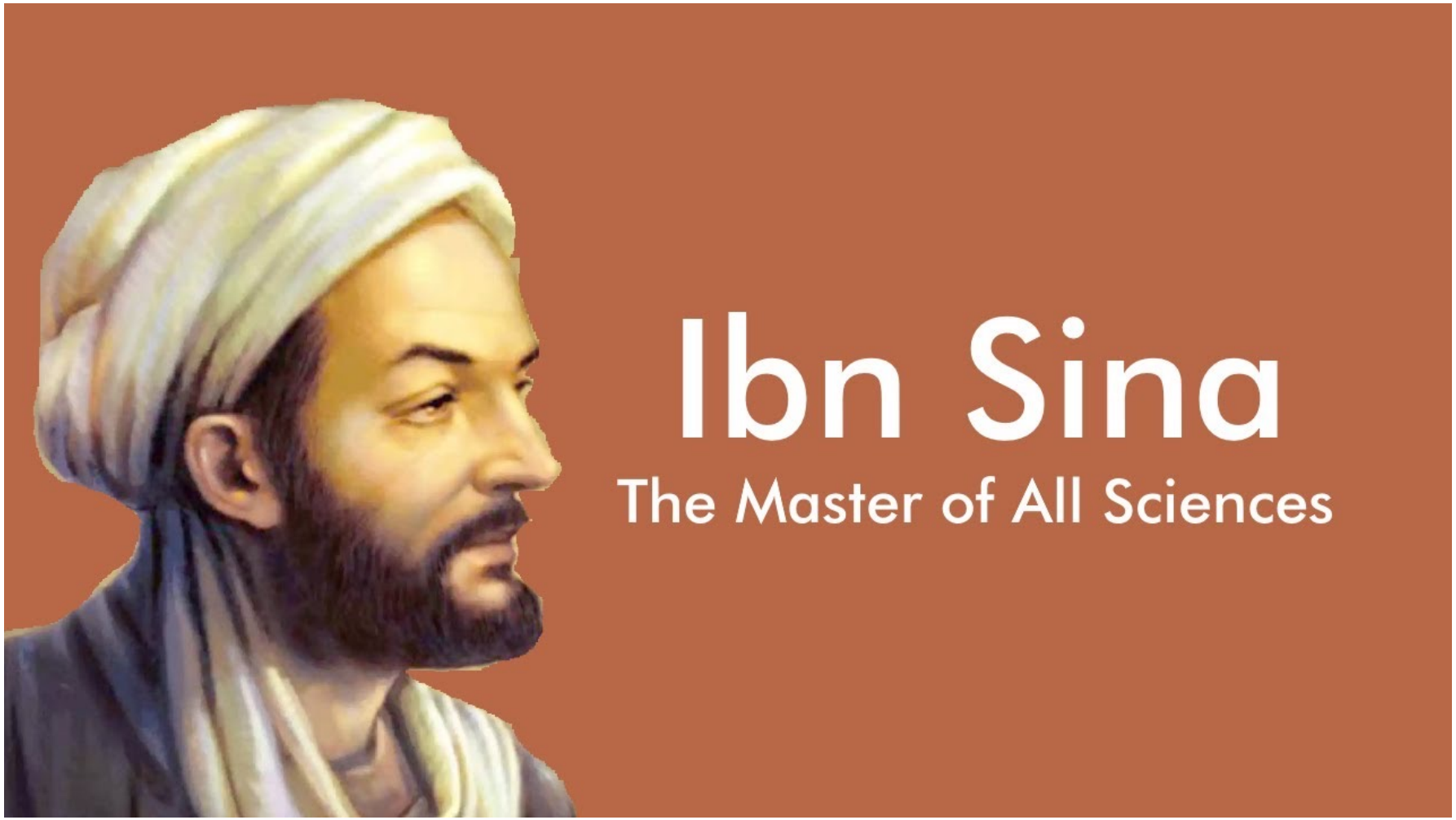
Avicenna- a great Islamic philosopher physician

# History of Biology

- During the post-classical era, **Middle Eastern medicine was the most advanced in the world**, integrating concepts of ancient Greek, Roman, Mesopotamian, Persian and the ancient Indian tradition of Ayurveda.

- **Ibn Sina, commonly known in the West as Avicenna**, was the preeminent philosopher and physician of the Muslim world.

- He is often described as the **father of early modern medicine**.




**Avicenna-a great Islamic philosopher physician**

# History of Biology

- During the **European Renaissance** and early modern period, biological thought was revolutionized in Europe and the discovery of many novel organisms.

- **Vesalius and Harvey**, who used experimentation and careful observation in physiology.

- **Andreas Vesalius** was an **anatomist and physician** who wrote one of the most influential books on human anatomy and a major advance over the long-dominant work of Galen.

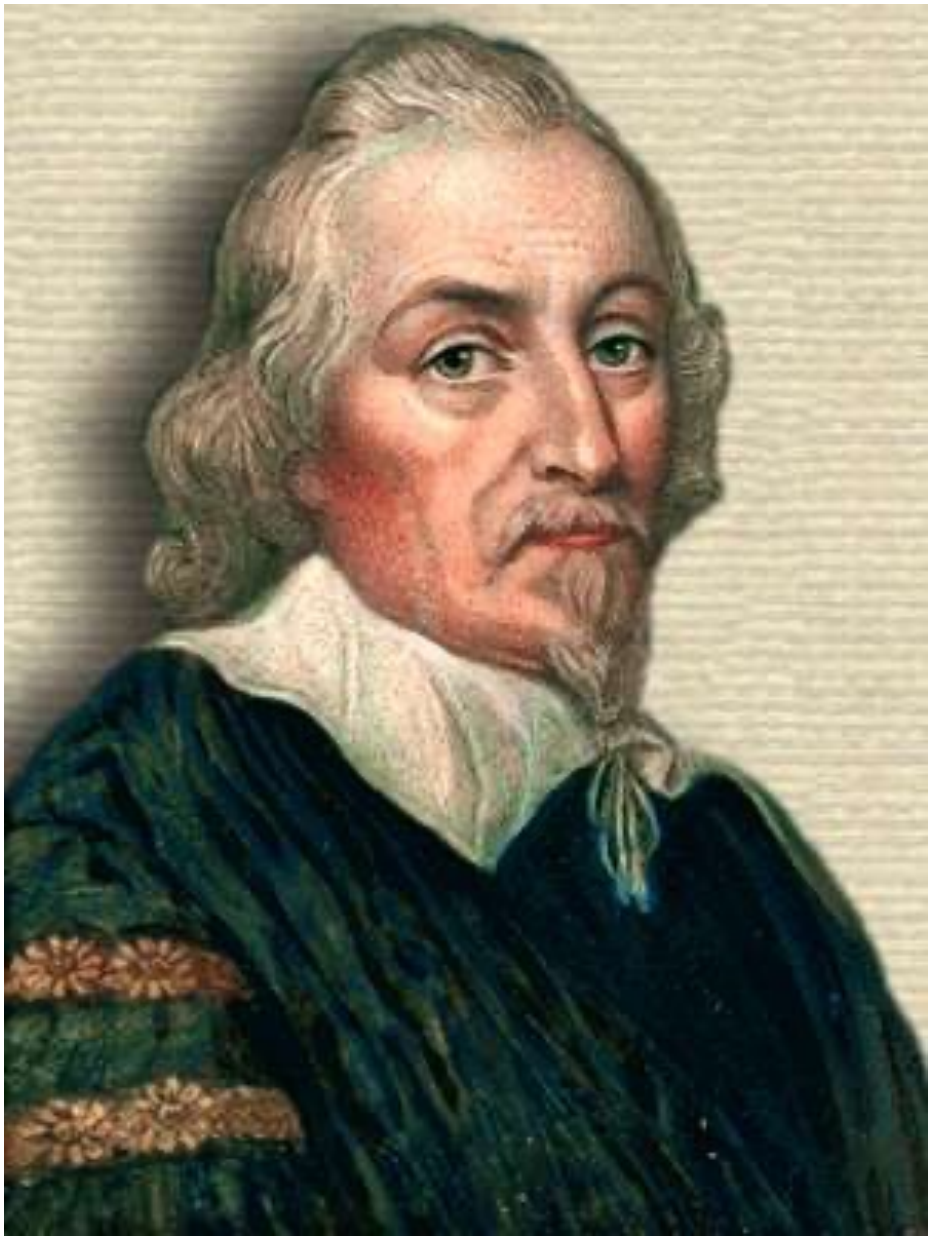
- Vesalius is often referred to as the **founder of modern human anatomy.**

Andreas Vesalius-founder of modern human anatomy

# History of Biology

- **William Harvey** was an English physician who made influential contributions in anatomy and physiology.

- He was the first known physician to describe completely, and in detail, the **systemic circulation** and properties of blood being pumped to the brain and the rest of the body by the heart.



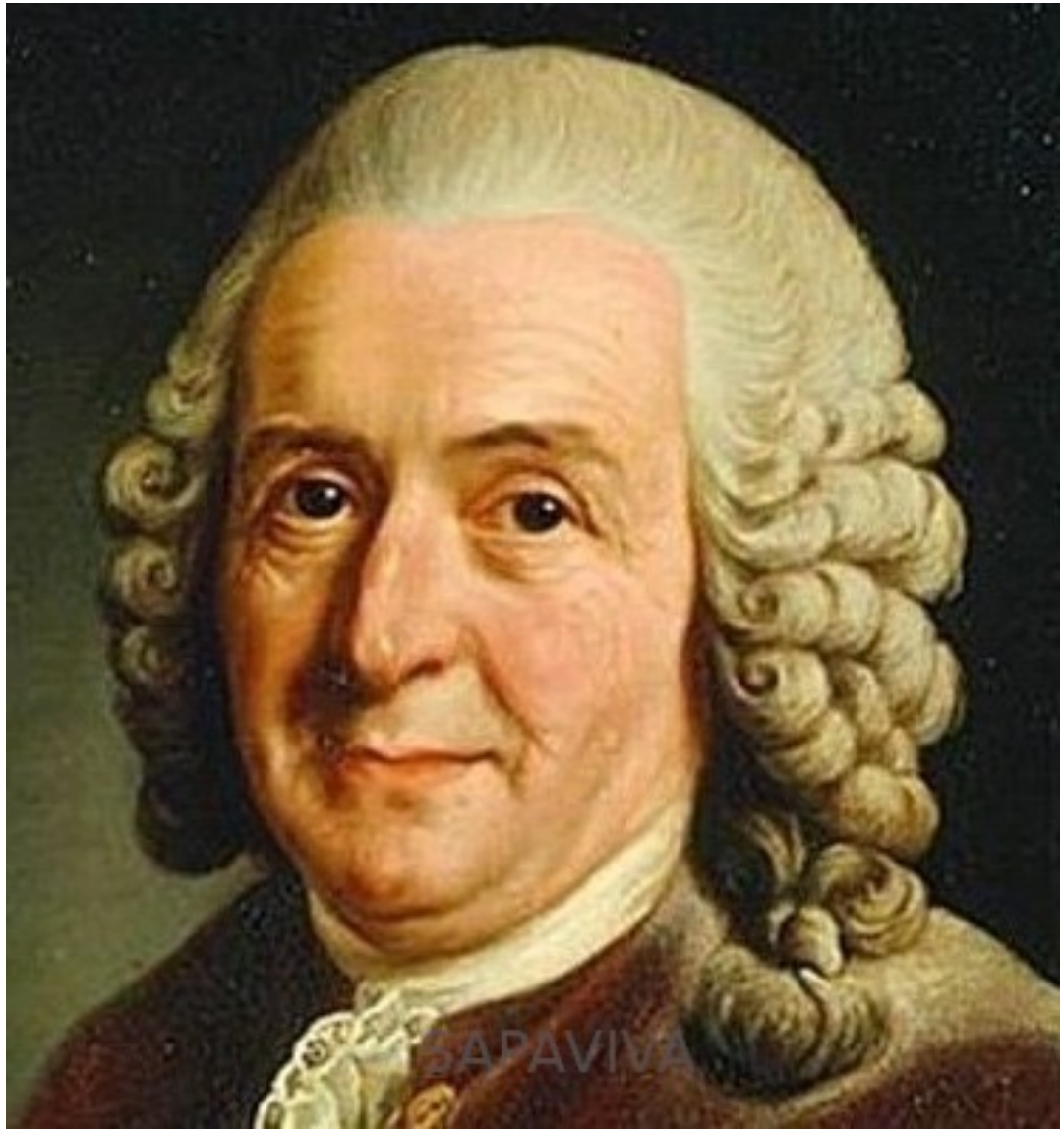
**William Harvey –Human blood circulation**

# History of Biology

- Naturalists such as **Linnaeus and Buffon** who began to classify the diversity of life and the fossil record, as well as the development and behavior of organisms.

- **Carl Linnaeus** was a **Swedish botanist, zoologist, taxonomist, and physician** who formalised binomial nomenclature, the modern system of naming organisms.

- He is known as the "**father of modern taxonomy**.




**Carl Linnaeus–father of modern taxonomy**

# History of Biology

- Over the **18th and 19th centuries**, biological sciences such as **botany and zoology** became increasingly professional scientific disciplines.

- **Lavoisier and other physical scientists** began to connect the animate and inanimate worlds through physics and chemistry.

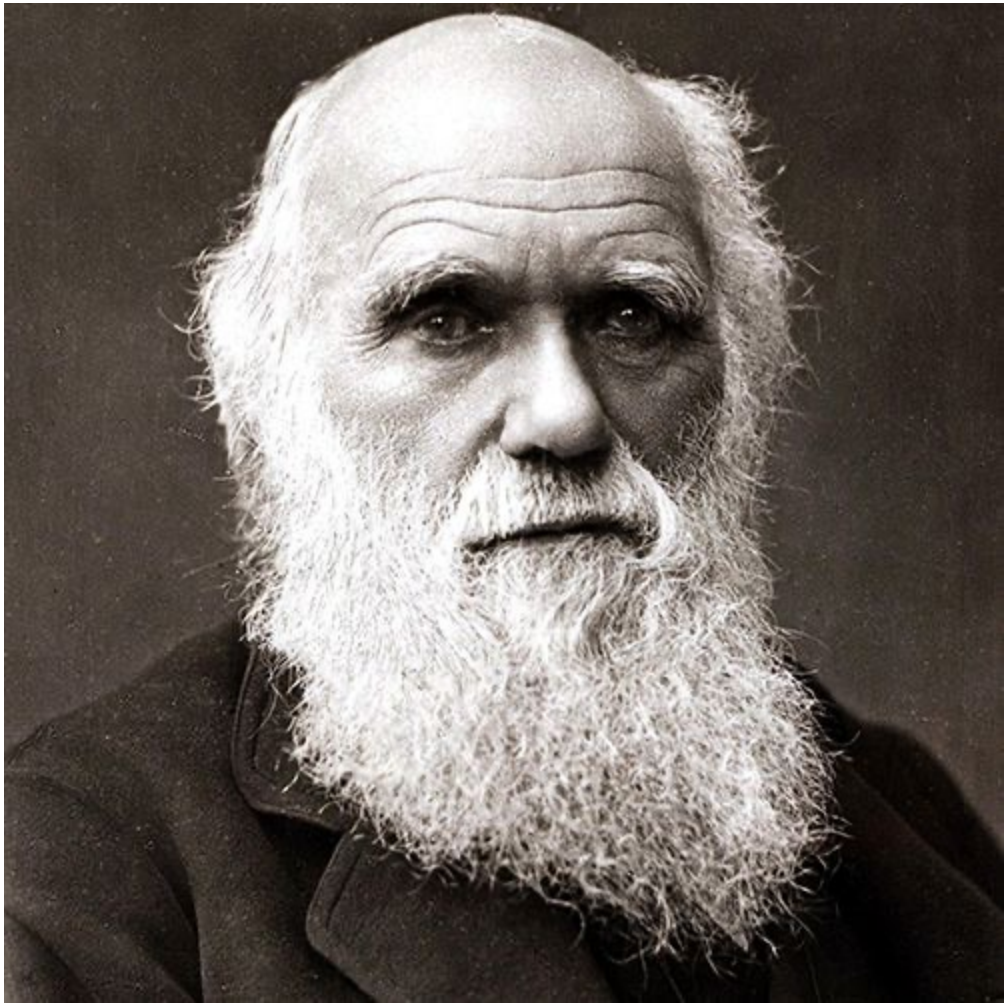
- Explorer-naturalists such as **Alexander von Humboldt** investigated the **interaction between organisms and their environment-**laying the foundations for biogeography and ecology.

**Alexander von Humboldt: organisms and their environment**

# The Humboldt University in Berlin, Germany

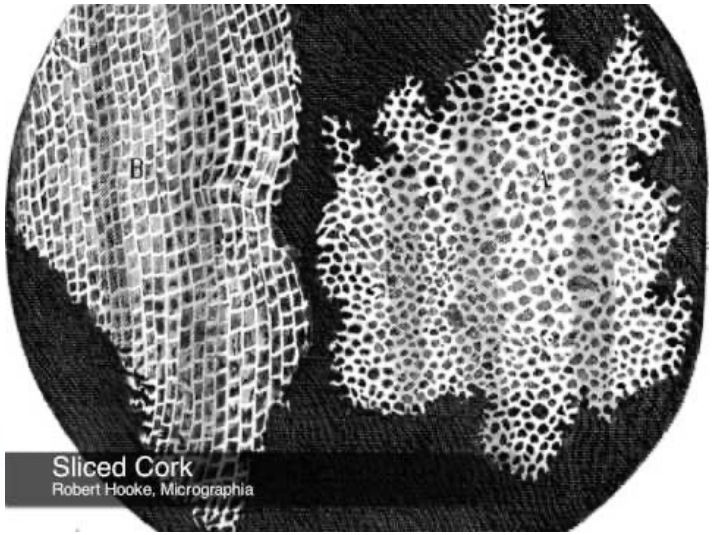
# History of Biology

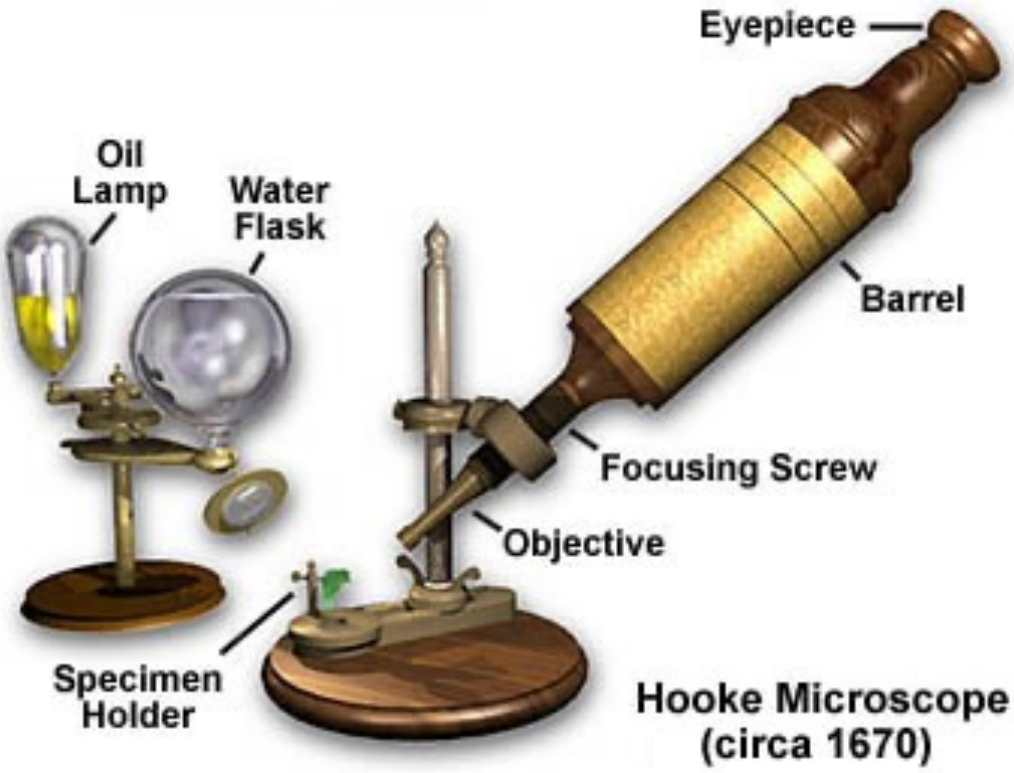
- **Charles Darwin's** theory synthesized the **evolution by natural selection**.



# History of Biology

- **Robert Hooke's** discovery marked the beginning of the cell theory the theory that all living things are composed of cells.

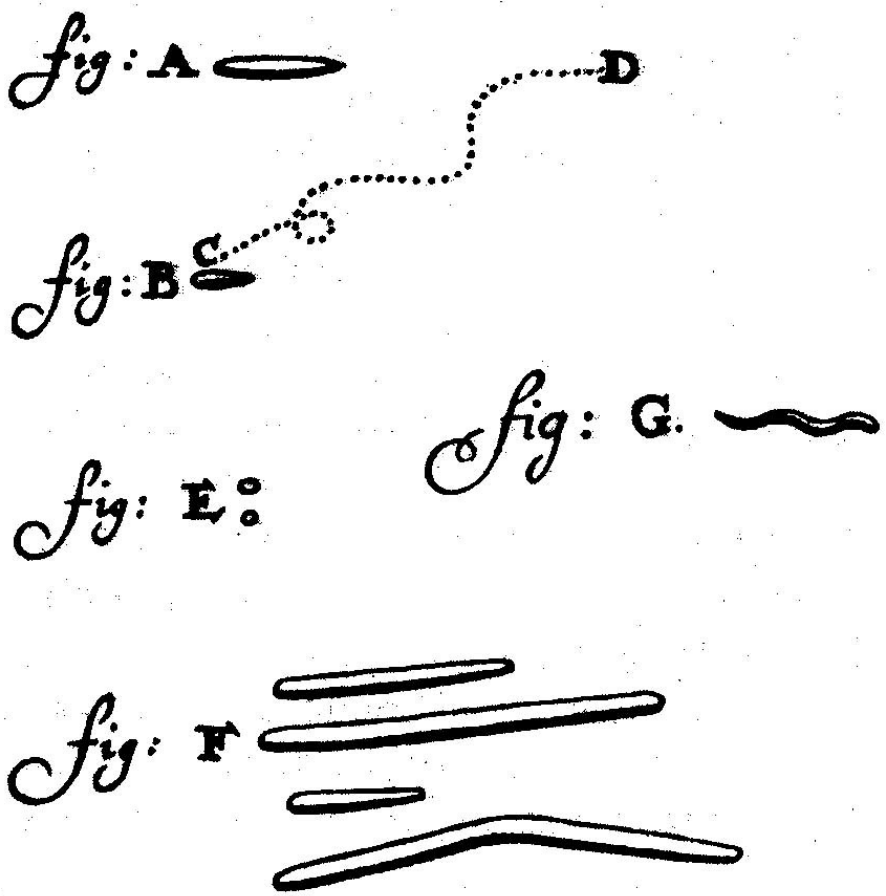


- **Antonie van Leeuwenhoek** revealed by means of microscopy the previously unknown world of microorganisms, laying the groundwork for cell theory.

- Antonie van Leeuwenhoek (1632–1723), known as the "father of microbiology," revolutionized biology by being the first to observe and describe single-celled organisms, which he termed "animalcules".

1632-1723



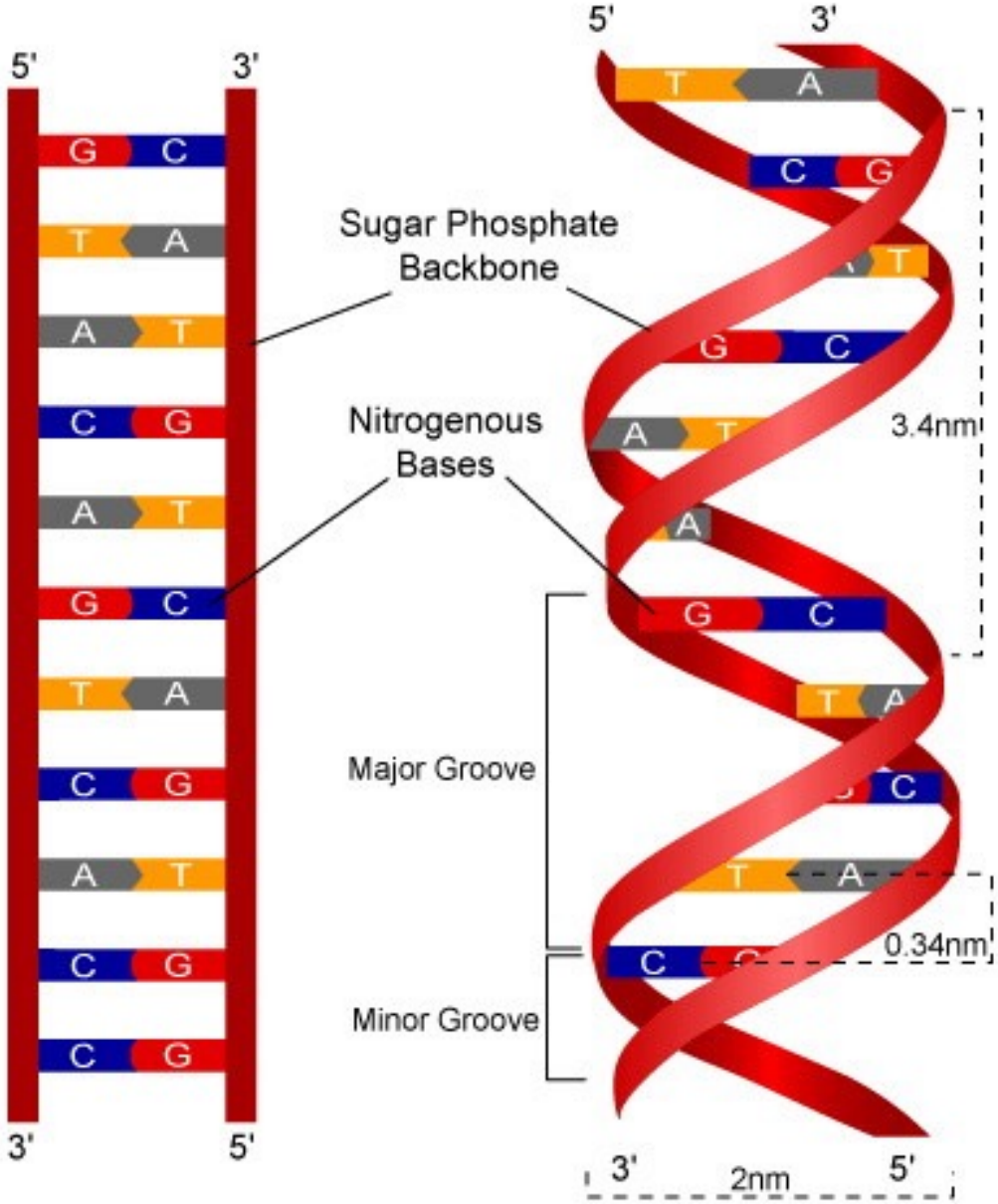
# History of Biology

- In the early 20th century, the rapid development of genetics was focused.

- In the mid of 20th century, **Watson and Crick** proposed the **structure of DNA** and new disciplines developed rapidly.

- By the late 20th century, new fields like **genomics and proteomics** have been established to find some signature makeup for individual organisms and populations.

- Further investigation is ongoing to investigate the interplay between **genes and the environment**.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Backbone
Nitrogenous
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Major Groove
Minor Groove
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**Watson and Crick: DNA structure**